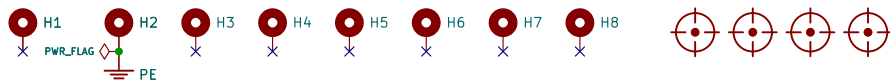

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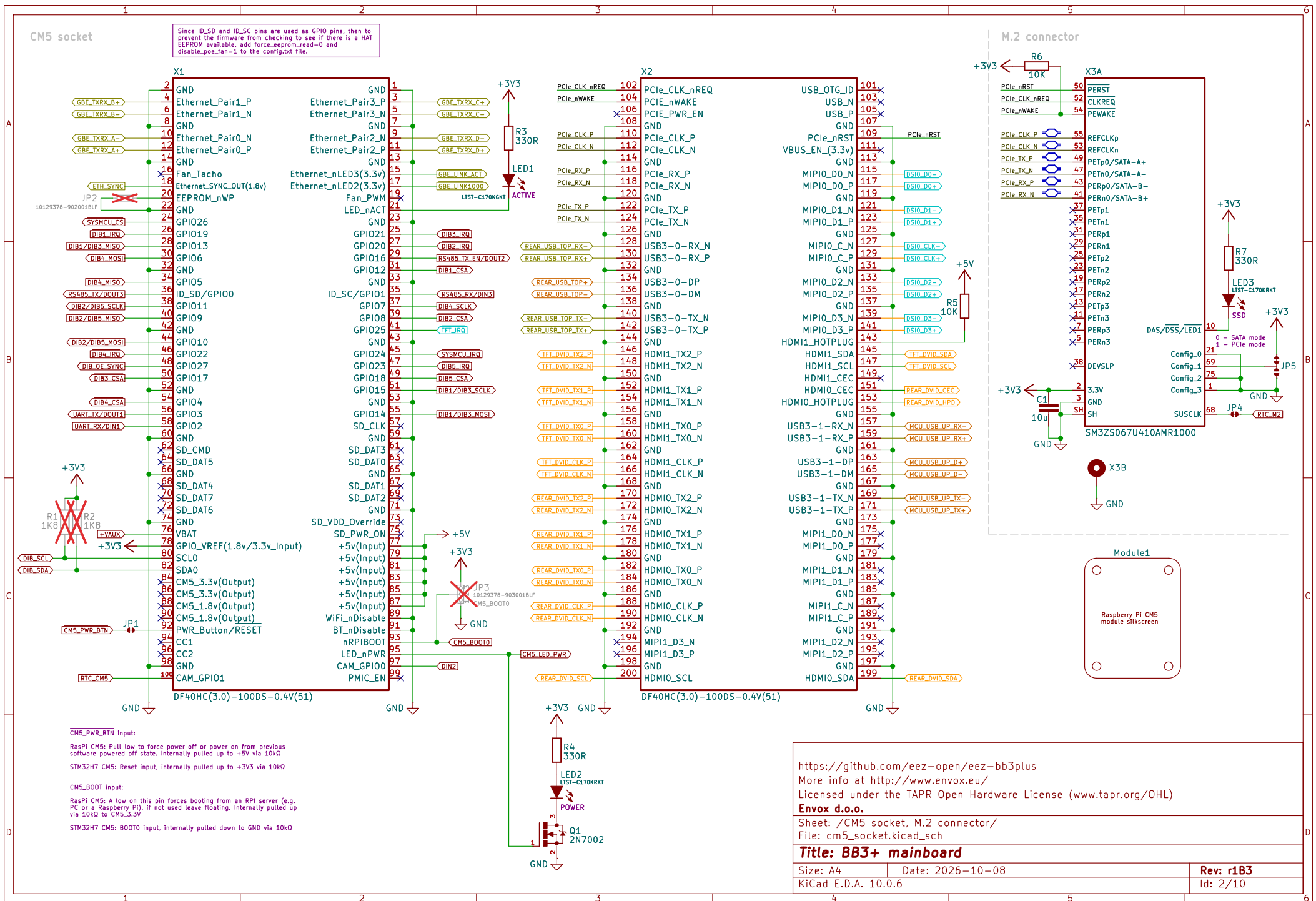
USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub PI Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

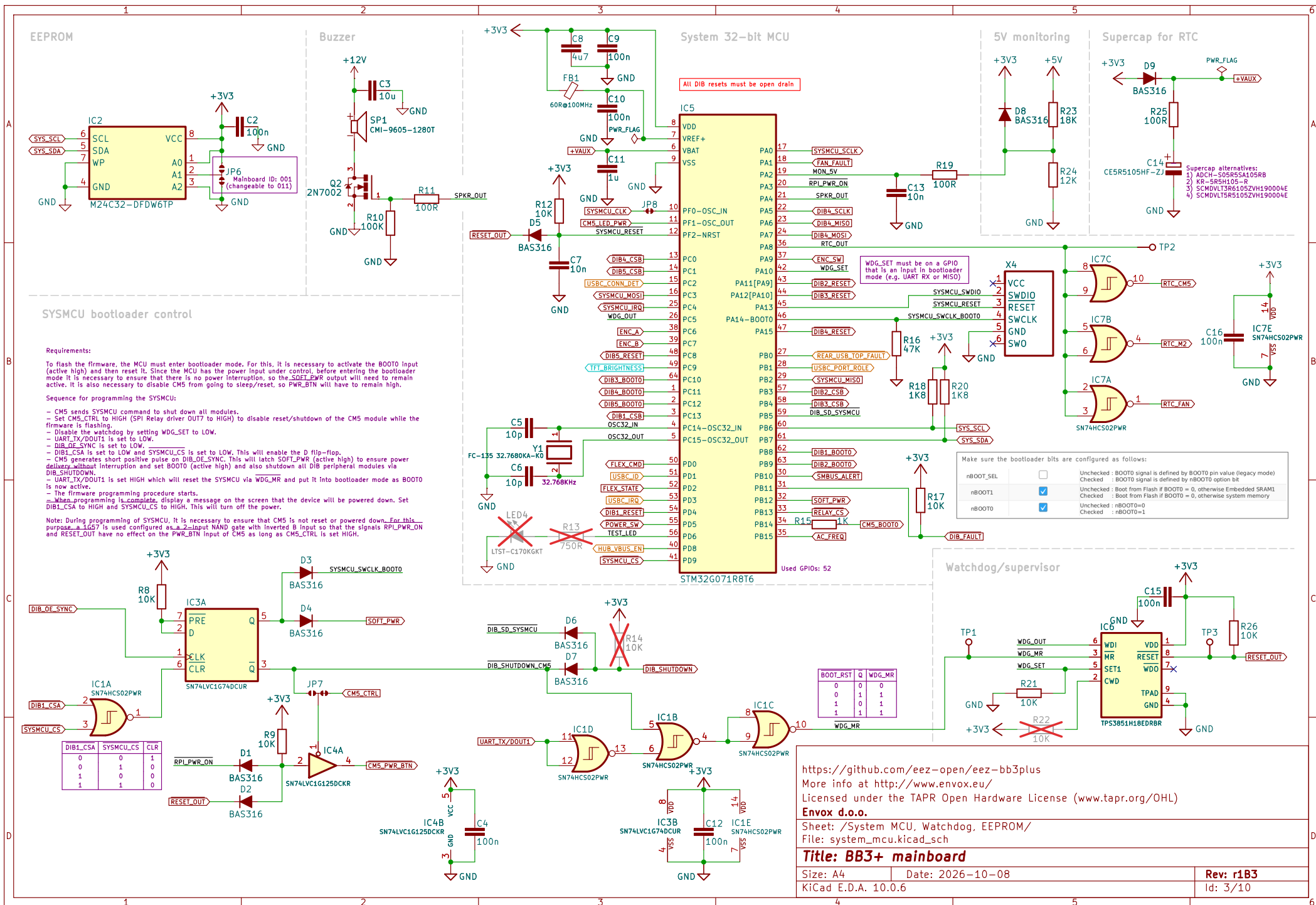
I2C address	Device
0x51	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x7A	PTN5150A USB CC controller



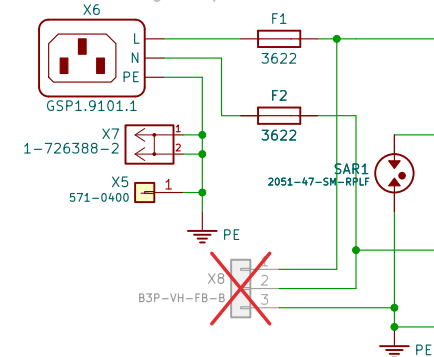
Title: BB3+ mainboard

Id: 1/10

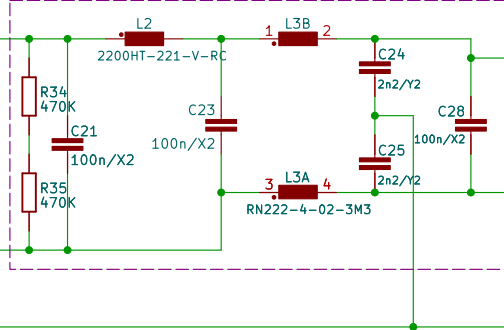




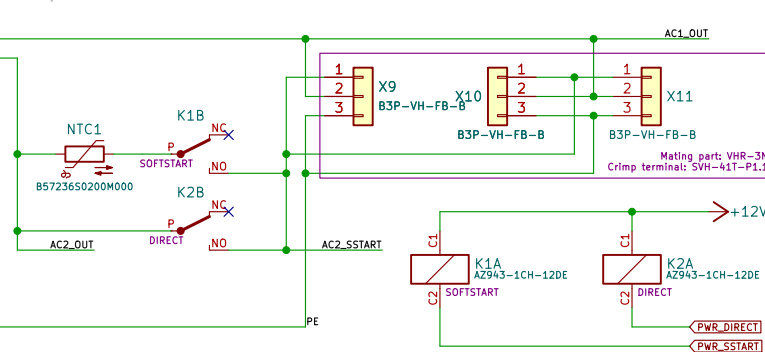
AC inlet, filtering and protection



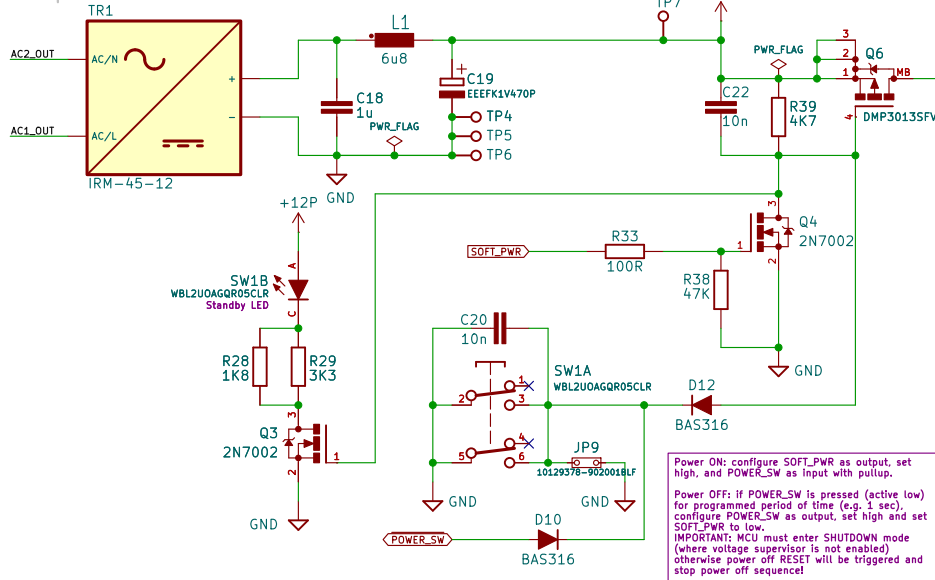
EMI compliance filter



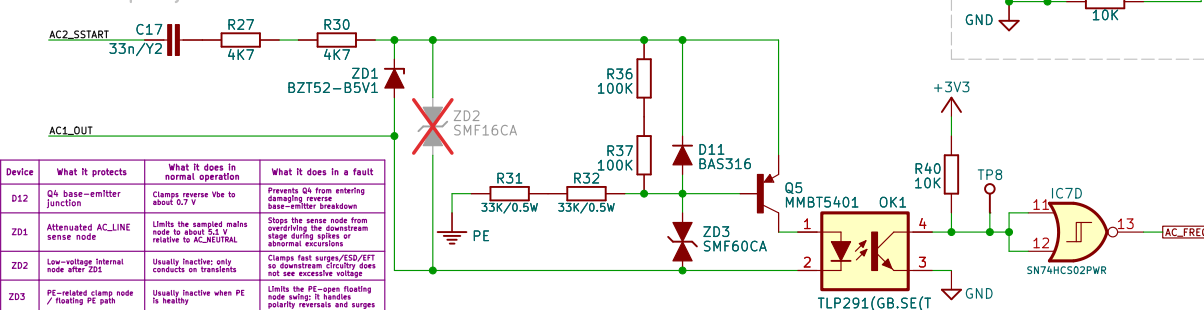
AC power control, Inrush current limiter



Soft power switch

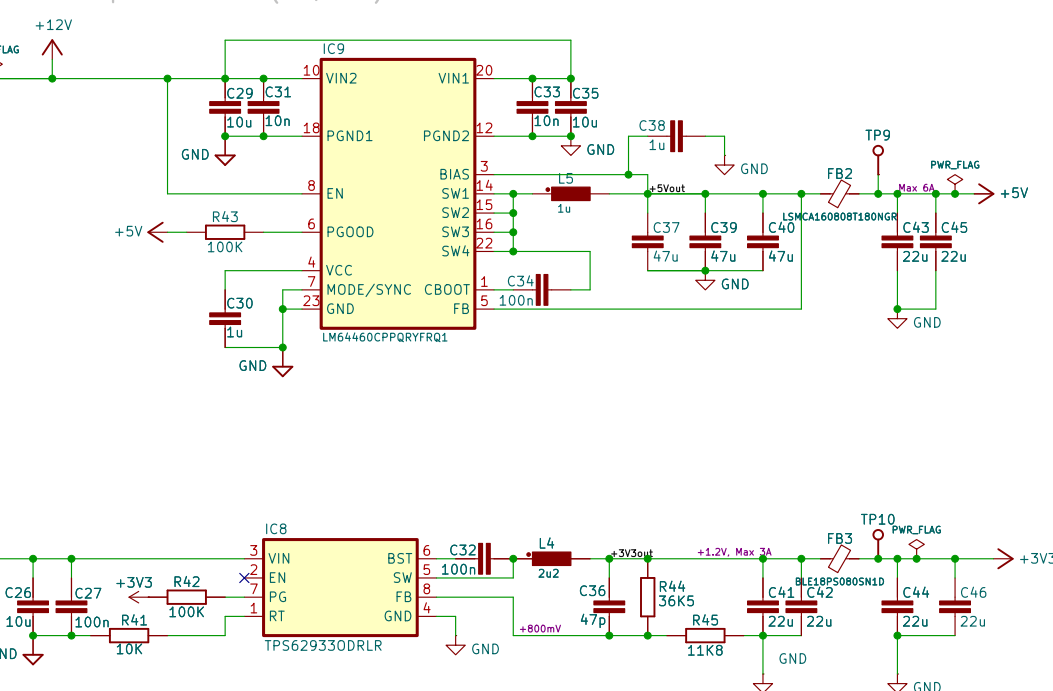


AC Mains frequency and PE fault detection



Device	What it protects	What it does in normal operation	What it does in a fault
D12	Q4 base-emitter junction		Prevents Q4 from entering damaging reverse base-emitter breakdown
ZD1	Attenuated AC_LINE sense node	Limits the sampled mains node to about 5.1V relative to AC_LINE05AL	Stops the sense node from overvoltage during spikes or abnormal excursions
ZD2	Low-voltage internal node after ZD1	Usually inactive; only conducts on transients	Clamps fast surges/ESD/RFI so downstream circuitry does not see excessive voltage
ZD3	PE-related clamp node / floating PE path	Usually inactive when PE is healthy	Limits the PE-open floating node swing; it handles polarity reversals and surges

DC-DC stepdown converters (+5V, +3V3)



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Sheet: /AC power, DC power, AC sync/

File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

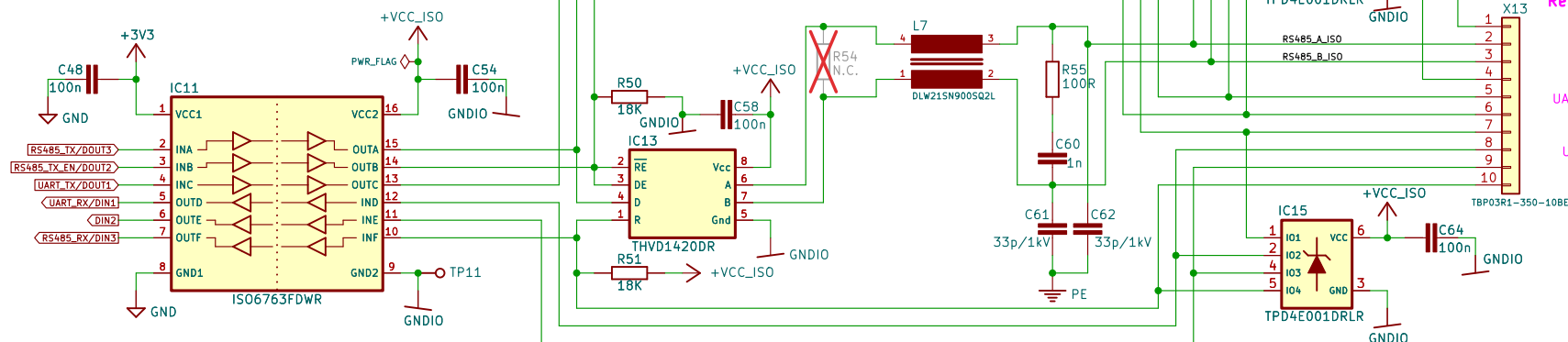
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 4/10

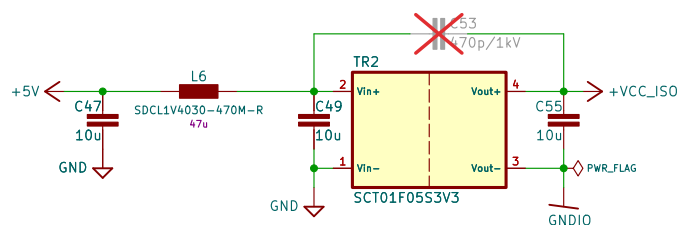
Isolated digital I/O & RS485 transceiver (half-duplex)



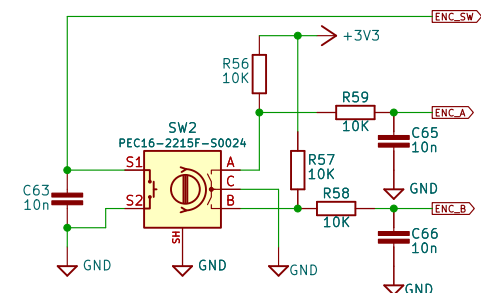
Rear panel I/O connector

+VCC_ISO	1
RS485_A_ISO	2
RS485_A_ISO	3
GND_ISO	4
UART_TX/DOUT1	5
DOUT2	6
DOUT3	7
UART_RX/DIN1	8
DIN2	9
DIN3	10

Isolated regulated power supply (+5V)



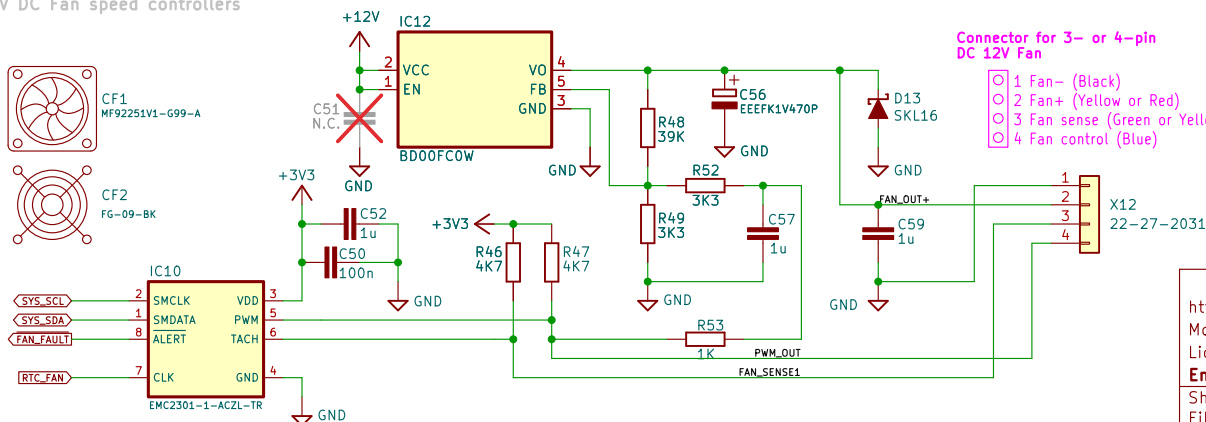
Encoder with switch



Encoder shaft length: 15 mm / Flatted

Knob head Ø31mm A1031648 (OKW) + Knob pointer A1101007 (OKW)
Optional: Knob head Ø31mm A3131068 (OKW) + Knob cap A3231007 (OKW)
Knob head Ø24mm A1024648 (OKW) + Knob pointer A1101007 (OKW)

12V DC Fan speed controllers



Connector for 3- or 4-pin DC 12V Fan

- 1 Fan- (Black)
- 2 Fan+ (Yellow or Red)
- 3 Fan sense (Green or Yellow)
- 4 Fan control (Blue)

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

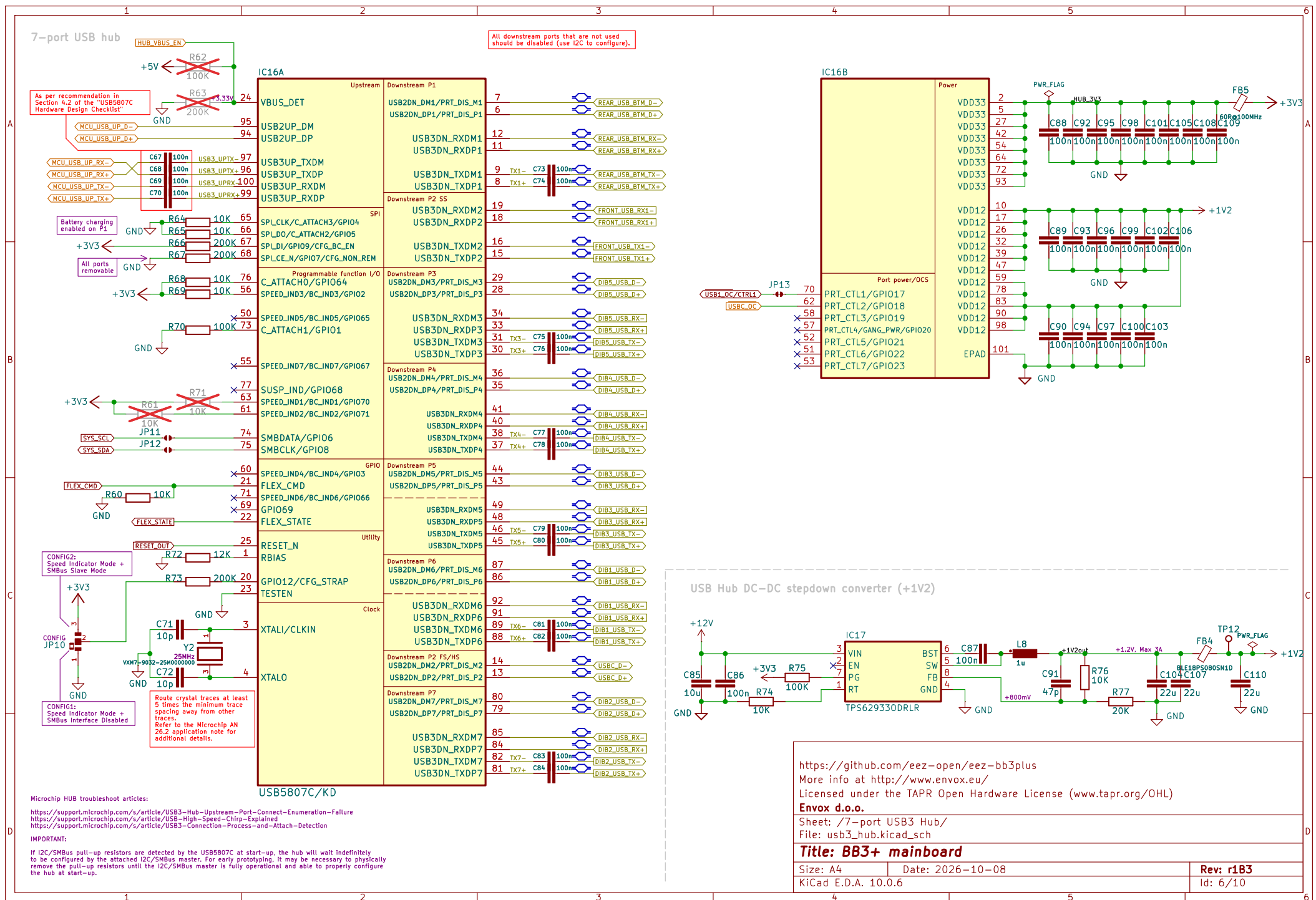
Title: BB3+ mainboard

Size: A4 Date: 2026-10-08

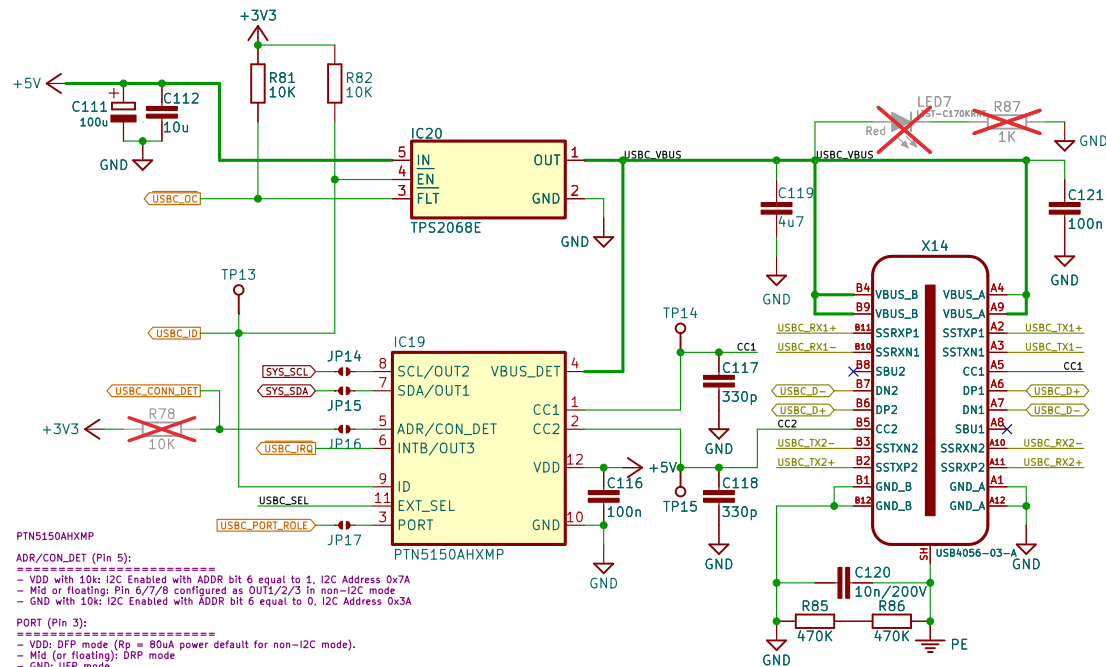
KiCad E.D.A. 10.0.6

Rev: r1B3

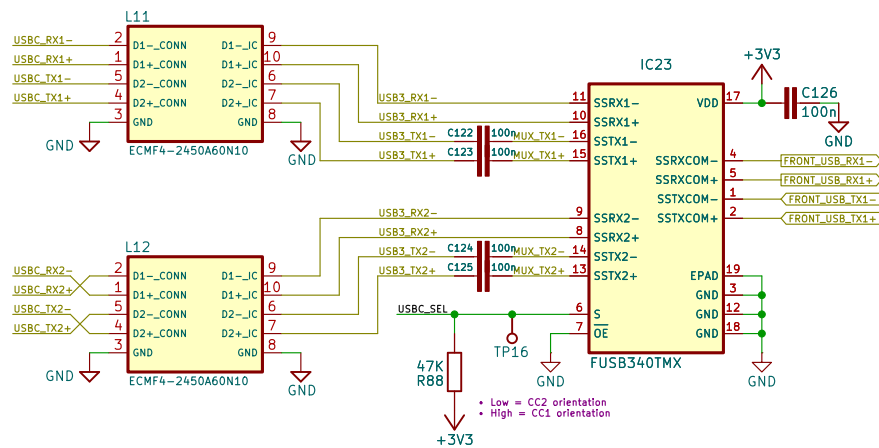
Id: 5/10



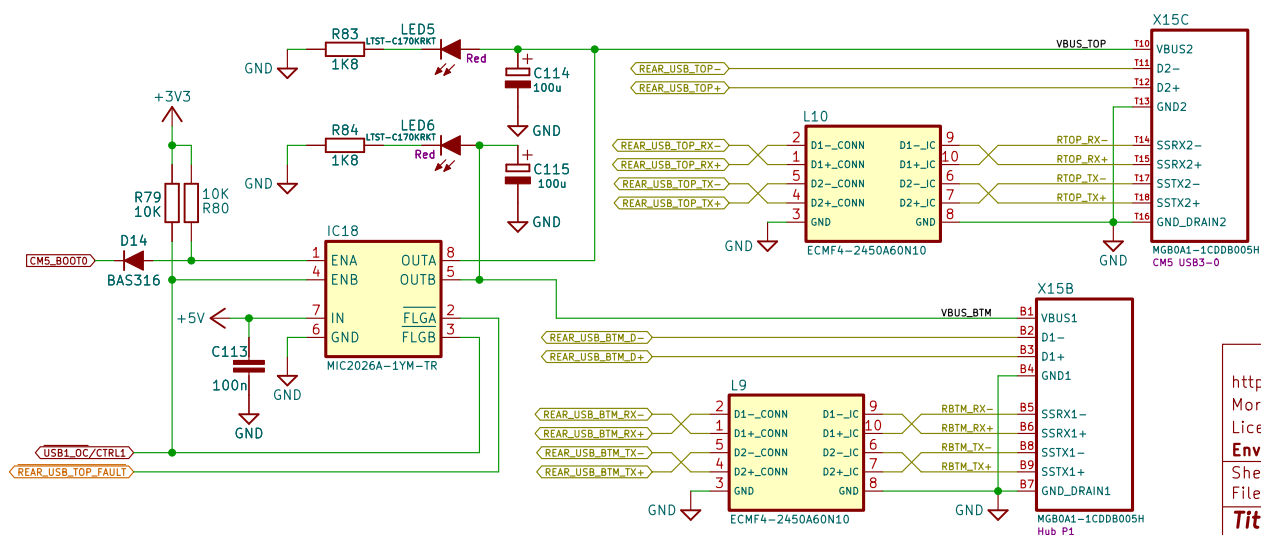
USB-C 3.0 front port (source max 1.5A)



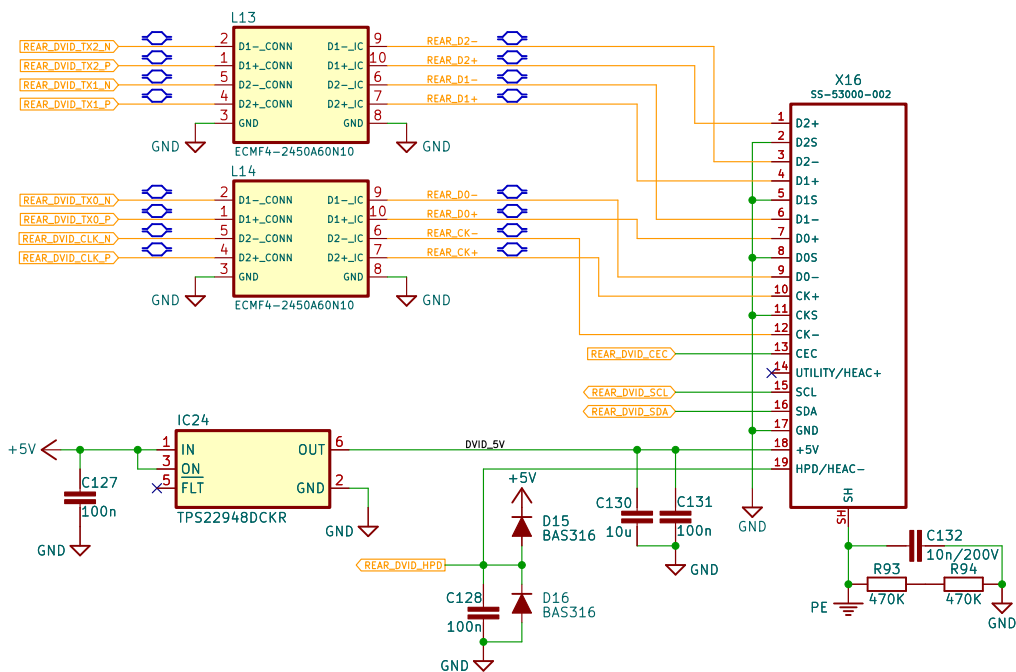
USB-C 3.0 Protection and Mux



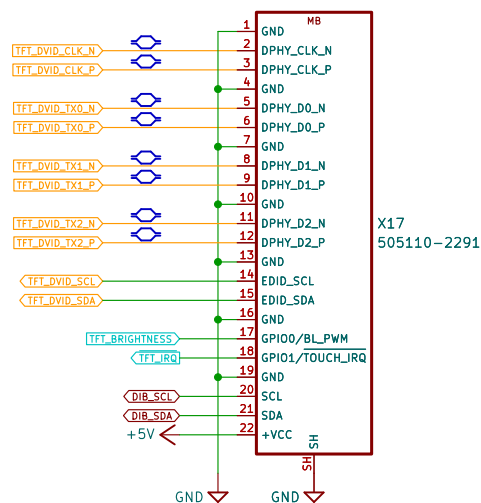
USB2.0 Type-A (Rear panel)



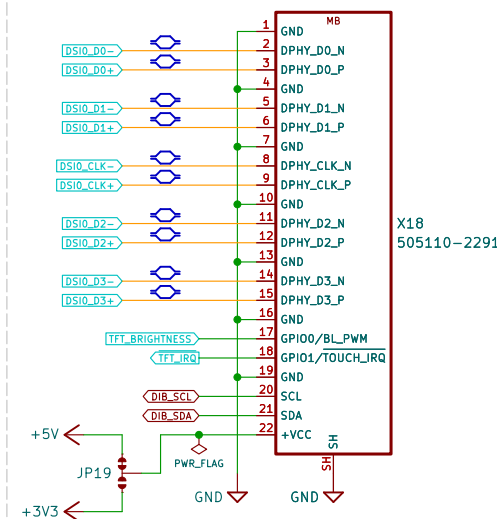
External monitor digital output



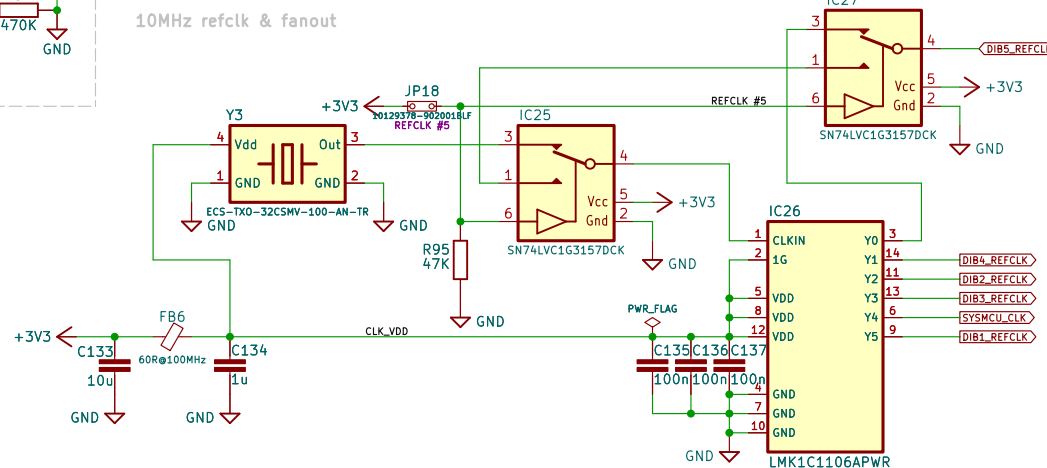
TFT Touchscreen display DVID output (22-pin FFC)



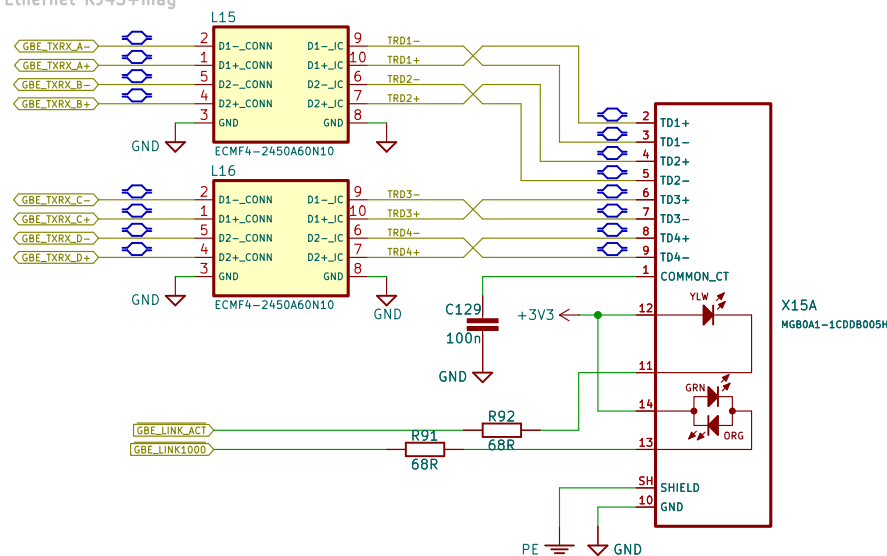
TFT Touchscreen display DSI output (22-pin FFC)



10MHz refclk & fanout



Ethernet RJ45+mag



<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: /Ethernet, Video outputs, Ref clk/

File: video_outputs.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.6

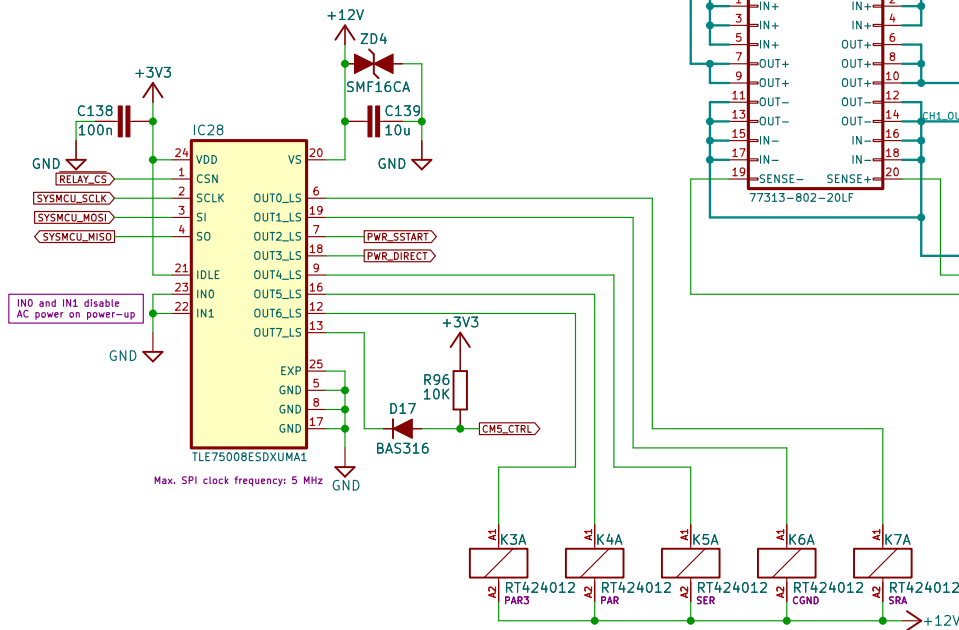
Rev: r1B3

Id: 8/10

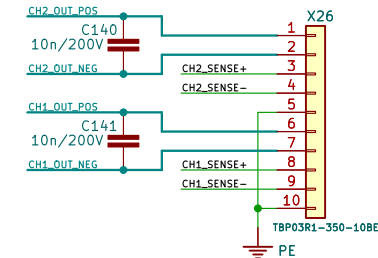
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	2
CH2_OUT- 2	3
CH2_SENSE+ 3	4
CH2_SENSE- 4	5
PE 5	6
CH1_OUT+ 6	7
CH1_OUT- 7	8
CH1_SENSE+ 8	9
CH1_SENSE- 9	10
PE 10	11

<https://github.com/eez-open/eez-bb3plus>

More info at <http://www.envox.eu/>

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Envox d.o.o.

Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.6

Rev: r1B3

Id: 10/10